CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2018 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is the difference between TCP and OSI Model?

Differences: OSI is a 7-layer theoretical conceptual model developed by ISO. TCP/IP is a 4-layer practical protocol suite developed by DoD that forms the basis of the modern Internet. OSI strictly separates session and presentation layers, whereas TCP/IP combines them into the Application layer.

Q2 (3M) What is Network Topology? Mention any two.

Topologies: Star topology (all nodes connect to a central hub/switch) and Bus topology (all devices share a single backbone cable terminated at both ends).

Q3 (3M) Explain Transmission Modes: Simplex, Half-Duplex, and Full-Duplex.

Modes: Simplex is strictly one-way (keyboard to monitor). Half-Duplex is bidirectional but one direction at a time (walkie-talkie). Full-Duplex is simultaneous two-way transmission (telephone conversation).

Q4 (3M) What is Modulation?

Definition: The process of varying one or more properties (amplitude, frequency, phase) of a high-frequency carrier wave with a data-bearing modulating signal for long-distance transmission.

Q5 (3M) What is Bit Rate vs Baud Rate?

Definitions: Bit rate is the number of bits transmitted per second (bps). Baud rate is the number of signal unit changes (symbols) per second. \text{Bit Rate} = \text{Baud Rate} \times \text{Bits per Baud}.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain any one protocol used for flow control in noisy channel: Stop-and-Wait ARQ.

Stop-and-Wait ARQ: Sender sends frame with sequence number (0 or 1) and starts timer. If positive ACK received, next frame is sent. If ACK lost or timer expires, frame is retransmitted. Handles lost frames and duplicate frames using 1-bit sequence numbering.

Q7 (7.5M) Explain Fiber Distributed Data Interface (FDDI) operation.

FDDI: High-speed 100 Mbps token-passing optical fiber network based on dual counter-rotating token rings (Primary and Secondary). If the primary ring breaks, the dual ring self-heals by wrapping around, ensuring fault-tolerant operation.

Q8 (7.5M) Explain Distance Vector Routing Algorithm.

Explanation: Routers periodically advertise cost estimates to all known destinations to immediate neighbors. Uses Bellman-Ford algorithm to compute least-cost paths.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain ISO/OSI Reference Model in detail with all seven layers and functions.**

OSI Model: Complete 7-layer functional architecture, services, and protocols.

Q10 (15M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat with comparisons.**

Protocols: Window sizing, ACK mechanism (cumulative vs individual), buffering requirements, and network utilization efficiency.

Q11 (15M) **Explain Error Detection using CRC (Cyclic Redundancy Check) with complete mathematical example.**

CRC Algorithm: Bit addition, modulo-2 division, remainder computation, and receiver verification.

Q12 (15M) **Explain Guided Media: Twisted Pair, Coaxial Cable, and Optical Fiber with advantages and limitations.**

Guided Media: Full physical characteristics, bandwidth capacity, attenuation rates, and shielding.

Q13 (15M) **Explain Circuit Switching, Packet Switching, and Message Switching with detailed comparison.**

Switching Methods: Complete comparison of connection setup, transmission delays, and resource utilization.
